

OFFICIAL LABORATORY TEST & COMPLIANCE CERTIFICATE

WIND LOAD RESISTANCE & STRUCTURAL DEFLECTION CERTIFICATE

Classification Class C5 (Maximum Design Pressure P1 = 2000 Pa / Safety P3 = 3000 Pa)

Governing Standard:	EN 12211:2016 (Method) • EN 12210:2016 (Classification)	Issuance Date:	January 15, 2026
Accredited Testing Facility:	EuroVent Architectural Facade Labs (Notified Body)	Audit Cycle:	2026 – 2029 (Triennial Review)
Manufacturer & Facility:	AinZara Aluminum Factory, Industrial Area, Tripoli	Specimen Origin:	Alloy 6063-T6 / PA66 Polyamide

1. SCOPE OF CERTIFICATION & TEST METHODOLOGY

Structural facade resistance testing evaluating frontal deflection under design wind load (P1), cyclic pressure pulsing durability (P2: 50 cycles of positive and negative gusting), and ultimate safety load resistance (P3 = 3000 Pa). Specimen integrity was monitored with electronic linear displacement transducers.

2. LABORATORY TEST RESULTS & CLASSIFICATION METRICS

Test Parameter / Clause	System / Substrate	Standard Target	Measured Value	Evaluation
Deflection Test @ P1 = 2000 Pa	CWA 50 SG Mullion (4.2m)	Relative Deflection < 1/300	Deflection = 1/412 (8.7 mm)	PASS (Class C)
Deflection Test @ P1 = 2000 Pa	SAT 120 Meeting Stile	Relative Deflection < 1/300	Deflection = 1/385 (6.2 mm)	PASS (Class C)
Cyclic Pressure Pulsing @ P2 = 1000 Pa	50 Repeated Cycles	No Mechanical Damage	Zero Displacement Drift	PASS (Class 5)
Safety Test @ P3 = 3000 Pa (+/-)	Full Frame Assembly	No Glass Dislodgement	No Permanent Strain	PASS (Integrity Preserved)
Post-Wind Air Permeability Repeat	WAT 63 Casement	Infiltration < Class 4 Max	Re-verified at Class 4	PASS (Compliant)
Corner Crimp Shear Strength	Alloy 6063-T6 Joints	Minimum 12.5 kN Force	Ultimate Failure > 21.4 kN	PASS (Exceeds Spec)

3. REGULATORY COMPLIANCE & USAGE ATTESTATION

The tested systems exhibited high structural rigidity with deflection staying well below the allowable L/300 threshold. Testing at 3000 Pa (equivalent to 252 km/h hurricane gusts) produced no hardware detachment, seal tear, or glazing failure. Approved for coastal, exposed hilltop, and supertall building elevations.

**ACCREDITED
DIRECTOR**

Dr. Eng. Marcus Von Berg, Ph.D.
Lead Auditor, Facade Testing Institute
Munich / EuroVent Network

LABORATORY

[SEAL]

**OFFICIAL SEAL OF CONFORMANCE
AIN ZARA TESTING BUREAU**

**DIRECTOR OF QUALITY
ASSURANCE**

Eng. Tarek Ben Mahmoud
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